Pranav M

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Research Interests: Processing-Near-Memory / Near-Memory Computing, Memory System Architecture, DDR/Memory Interfaces, CPU & SoC Design

Education

2023–2027 B.Tech, Electronics and Communication Engineering, **PES University**
2022 Class 12th, **Vydehi School of Excellence**

Research Experience

HADES — Programmable Processing-Near-Memory Engine 2026
Independent Research Project GitHub arXiv

Designed and verified a programmable processing-near-memory engine for binary similarity search. Developed a 12-opcode, 16-bit ISA with dual-row scan and selective-scan mechanisms around a shared, unmodified 2R1W register-file macro. Submitted for tapeout through the Tiny Tapeout SKY130 flow. Post-layout implementation closes timing at 40 MHz with zero reported DRC, LVS, and antenna violations. Performance comparisons are based on pre-silicon simulation.

Aletheia — Processing-Near-Memory Framework 2026
Independent Research Project GitHub

Built an experimental PNM framework using a Raspberry Pi 5 as a memory node that stores datasets locally and executes workloads without transferring the underlying data to the host. Implemented TCP-based command dispatch and PMU-backed memory characterization across working-set sizes, stride patterns, and dataset scales to investigate cache behavior, memory locality, and access costs.

Publication

HADES: A Programmable Processing-Near-Memory Engine for Binary Similarity Search
arXiv preprint, 2026 arXiv:2610.14677

Projects

Reconfigurable DDR5 Memory Controller (RMC) GitHub

Architecting a parameterized, DDR5-first memory controller targeting JEDEC JESD79-5 and DFI 5.2, with an AXI4 host interface and scalable multi-channel/multi-rank support. Developed the client-interface and transaction-management architecture, including address validation and mapping, burst splitting, per-AXI-ID ordering, and read-path verification. Specified the MC-core datapath and scheduling interfaces, APB/CSR register map, DDR5 mode-register programming, and power-management flows. Contributed a DFT observation interface and read-path testbenches. MC-core RTL integration and end-to-end evaluation remain ongoing.

Specula — Out-of-Order RISC-V Core GitHub

Implemented a speculative, single-issue out-of-order RISC-V core with physical-register renaming, a 32-entry reorder buffer with in-order commit, an 8-entry reservation station, and gshare/BTB branch prediction with misprediction recovery. Developed a decoupled memory pipeline with an 8-entry store queue and byte-lane store-to-load forwarding; integrated machine/supervisor privilege modes, Sv32 translation, and simulated UART, PLIC, and VirtIO block device support. Five standalone benches and 16 directed pro-

grams passed in Bluesim; xv6-rv32 reached an interactive shell in simulation. Vivado synthesis reported +0.996 ns WNS at a 90 MHz constraint.

Herald — Fixed-Point DSP Coprocessor GitHub

Designed and taped out a fixed-point Q12.12 DSP coprocessor for FPU-less microcontrollers through the Tiny Tapeout IHP SG13G2 130 nm shuttle. Features a CORDIC engine for trigonometric and vector operations and a multiply-accumulate (MAC) unit. Verified using Bluespec and cocotb. Physical-design results include 50 MHz operation, approximately 6 ns worst-case setup slack, zero routing overflow, and approximately 80,000 μm^2 area at 62.9% utilization.

Aetheron — RISC-V SoC GitHub

Built a simulation-based RISC-V SoC in Bluespec SystemVerilog, running compiled C programs. Includes a pipelined RV32I core, boot-from-ROM loading of C payloads into RAM, a TileLink-lite interconnect, and memory-mapped UART, GPIO, and timer peripherals. Developed the supporting C/assembly toolchain and custom linker script.

Teaching & Mentorship

PES University — Teaching Assistant, Computer Organization & Design Aug 2026–Present

Teach RISC-V assembly and processor microarchitecture, and guide students through the design, implementation, simulation, and debugging of single-cycle RISC-V processors.

HSP, PES University — Mentor Jul 2025–Present

Mentored students on digital hardware design projects across two consecutive terms, guiding architecture design, reviewing HDL, and helping debug design modules.

Skills

Languages	SystemVerilog, Verilog, Bluespec, Veryl, C, Python, Assembly
Tools & Platforms	Vivado, GTKWave, Verilator, Icarus Verilog, cocotb, ModelSim, Git, Linux
Domains	Computer Architecture, Memory Systems, DDR/Memory Interfaces, Processing-Near-Memory, RTL Design and Verification, SoC Design

Writing

Maintains a research-focused blog and weekly TIL/research-diary practice at prawns.dev, documenting ongoing work in computer architecture and memory systems.